

Software Requirements Specification (SRS)

Project: Quantum Bluff

Version 1.0 - Milestone M1 Validation

Project Leader: Omar

Tech Lead: Mohamed

QA/Analyst: Linda

Scientific Advisor: Cyrille Baudrier

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0.1 Introduction

0.1.1 Purpose

The purpose of this document is to define the functional and technical specifications for the **Quantum Bluff** desktop application. This document serves as the primary reference for development, quality assurance, and project validation.

0.1.2 Project Scope

Quantum Bluff is a high-stakes strategy game involving complex probability mechanics (*Quantum Speculation*). The software is developed as a ****100% local desktop client****, prioritizing mathematical integrity and real-time performance without the need for an internet connection.

0.1.3 Definitions and Roles

- **PL (Omar):** Global oversight and strategic validation.
- **Tech Lead (Mohamed):** Technical architecture and code integrity.
- **QA (Linda):** Validation of requirements and regression testing.
- **Scientific Lead (C. Baudrier):** Modeling of the Hidden Bets algorithms.

0.2 Overall Description

0.2.1 Product Perspective

The application is a standalone Java-based software using JavaFX for the user interface. It operates entirely on the user's machine, ensuring data privacy and offline accessibility.

0.2.2 Operational Environment

- **Operating System:** Windows 10/11, macOS, Linux.
- **Runtime:** Java Runtime Environment (JRE) 17 or higher.
- **Connectivity:** None (100% Offline).

0.3 Functional Requirements

0.3.1 Core Game Mechanics

FR-1 Local Session Management: The system shall allow users to start, pause, and save game sessions locally without account creation.

FR-2 Hidden Bets Engine: The system shall implement the "Quantum Speculation" mechanic where players bet on hidden probabilistic outcomes.

FR-3 Odds Calculation: The system shall calculate real-time quotas using the validated *Baudrier* formulas.

FR-4 Pass-and-Play: Support for local multiplayer through a single terminal interface.

0.3.2 Data Persistence

FR-5 Local Storage: Game states and statistics shall be stored in local JSON or SQLite files.

FR-6 Integrity Check: The system shall verify the checksum of local save files to prevent unauthorized manual editing of scores.

0.4 Technical Specifications

0.4.1 Technology Stack

- **Language:** Java 17.
- **UI Framework:** JavaFX (MVC Pattern).
- **Build Tool:** Maven (generating a Fat JAR via Maven Shade Plugin).
- **CI/CD:** GitLab CI (automated build and unit testing).

0.4.2 Architecture Requirements

- **Multithreading:** Separate threads for the probability engine and the UI to prevent interface freezing during complex calculations.
- **Modularity:** Decoupling of the game logic (*Azra/Souheil*) from the display components.

0.5 Validation and Quality Assurance

0.5.1 Validation Protocol (M1)

The specifications have been validated through a three-step process:

1. **Technical Review:** Conducted by Mohamed to ensure architectural feasibility.
2. **QA Review:** Conducted by Linda to ensure requirements traceability.
3. **Final Approval:** Signed by Omar on February 15, 2026.

0.5.2 Non-Functional Requirements

- **Performance:** Quota calculation must return results in less than 100ms.
- **Security:** Protection of the "Hidden Bets" logic from simple decompilation.